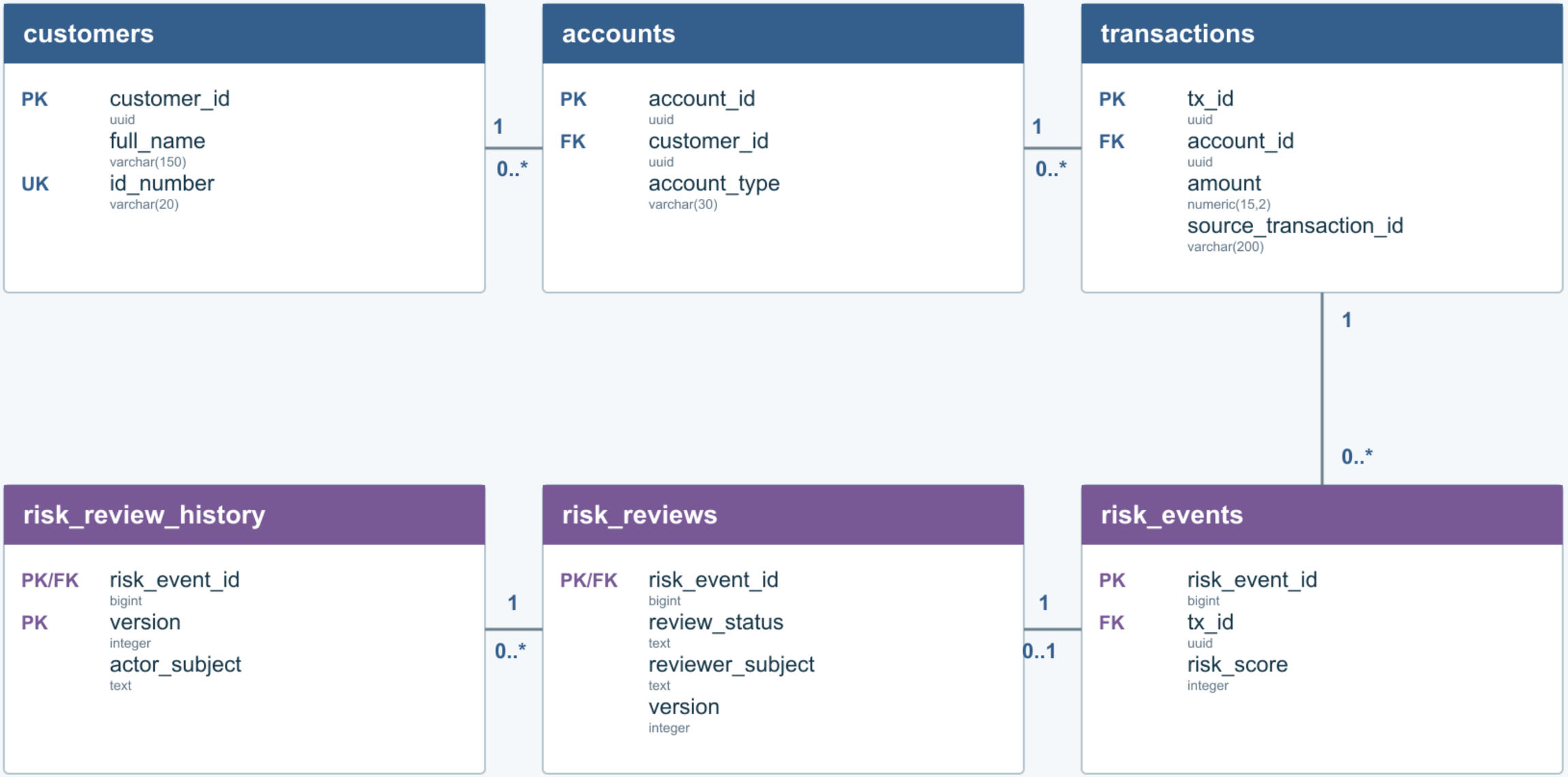
The transaction and its investigation

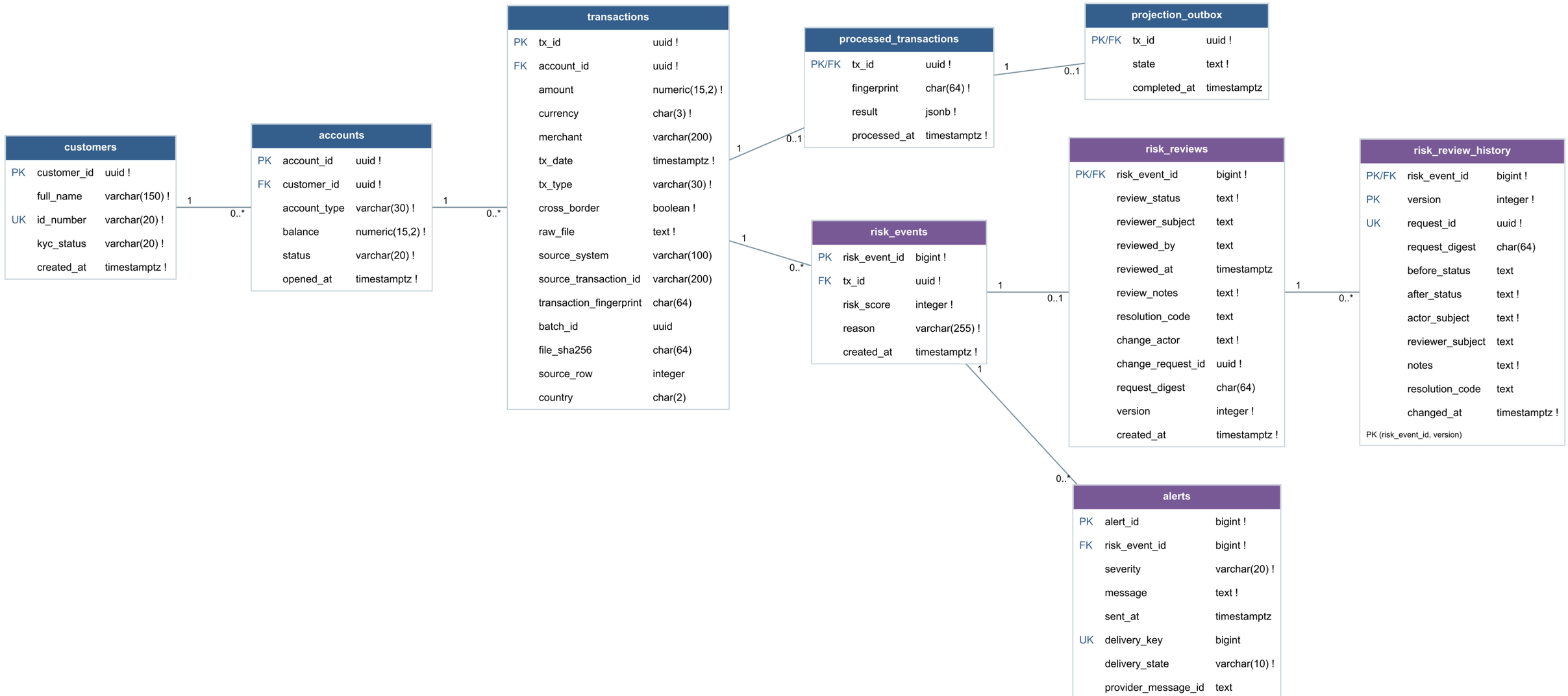
Current SQL names and key relationships. The full ERD contains all 18 tables.



PK: primary key. FK: foreign key. UK: unique key. Composite history key: (risk_event_id, version).

Selected columns only. Stored assessment details also live in processed_transactions.result (JSONB).

Ledger, frozen assessments and investigation | All columns
PK primary key FK foreign key UK unique key ! required column
Numbers describe database constraints, not process order



Ingestion, quarantine and correction | All columns
PK primary key FK foreign key UK unique key ! required column
Numbers describe database constraints, not process order

transactions

PK	tx_id	uuid !
FK	account_id	uuid !
	amount	numeric(15,2) !
	currency	char(3) !
	merchant	varchar(200)
	tx_date	timestamptz !
	tx_type	varchar(30) !
	cross_border	boolean !
	raw_file	text !
	source_system	varchar(100)
	source_transaction_id	varchar(200)
	transaction_fingerprint	char(64)
	batch_id	uuid
	file_sha256	char(64)
	source_row	integer
	country	char(2)

transaction_lifecycle_events

PK	event_id	bigint !
FK/UK	batch_id	uuid !
FK/UK	source_row	integer !
UK	sequence	integer !
	event_type	text !
	occurred_at	timestamptz !
	actor_subject	text !
	error_type	text

UK (batch_id, source_row, sequence)

ingestion_attempts

PK/UK	attempt_id	uuid !
FK/UK	batch_id	uuid
	source_bucket	text !
	source_file	text !
	source_version	text
	source_etag	text
	received_at	timestamptz !
	status	text !
	error_code	text

UK (batch_id, attempt_id)

ingestion_records

PK/FK	batch_id	uuid !
PK	source_row	integer !
FK	source_attempt_id	uuid !
	disposition	text !
	candidate_digest	char(64)
FK	canonical_tx_id	uuid
	reason_code	text
	updated_at	timestamptz !

PK (batch_id, source_row)

quarantine_cases

PK	quarantine_id	uuid !
FK/UK	batch_id	uuid !
FK/UK	source_row	integer !
	reason_code	text !
	reason_detail	text !
	validation_errors	jsonb !
	candidate_account_id	uuid
	safe_payload_key	text !
	safe_payload_version	text !
	source_reference	text
	quarantined_at	timestamptz !

UK (batch_id, source_row)

ingestion_batches

PK	batch_id	uuid !
	source_system	varchar(100) !
UK	source_bucket	text !
UK	file_sha256	char(64) !
	received_at	timestamptz !
	rows_received	integer
	publication_complete	boolean !
	file_failure_code	text

UK (source_bucket, file_sha256)

quarantine_corrections

PK	correction_id	uuid !
FK/UK	quarantine_id	uuid !
UK	request_id	uuid !
	request_digest	char(64)
UK	correction_version	integer !
	submitted_by	text !
	submitted_at	timestamptz !
	correction_notes	text !
	corrected_payload_key	text
	corrected_payload_version	text
	action	text !
FK	reprocess_batch_id	uuid
FK	reprocess_source_row	integer

UK (quarantine_id, correction_version)

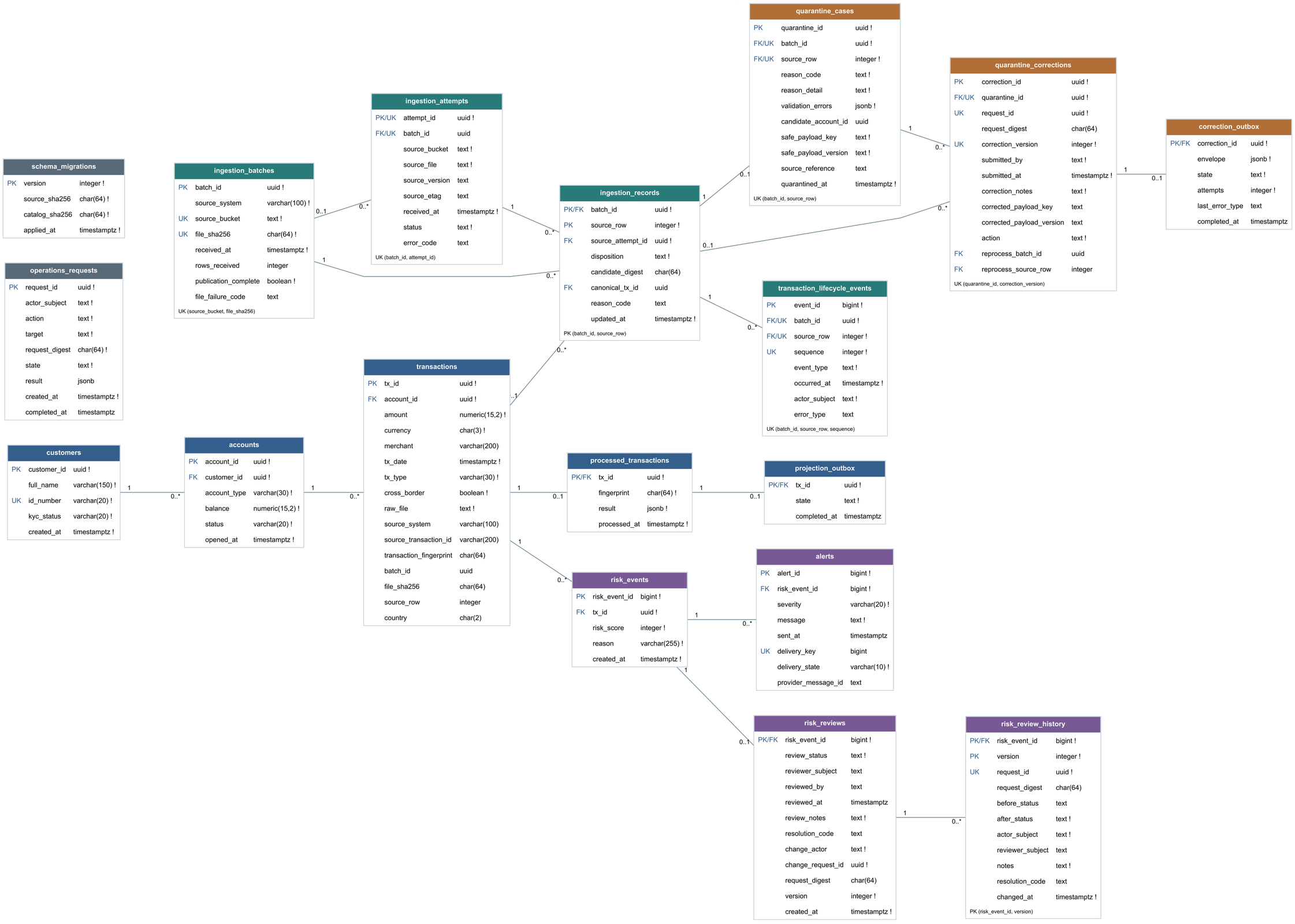
correction_outbox

PK/FK	correction_id	uuid !
	envelope	jsonb !
	state	text !
	attempts	integer !
	last_error_type	text
	completed_at	timestamptz

Imbokodo Risk and Compliance Workbench | Full physical ERD | 18 tables

PK primary key FK foreign key UK unique key ! required column

Numbers describe database constraints, not process order



Source and reading guide

18 PostgreSQL tables and 17 declared foreign keys. The two derived workflow views, batch_reconciliation and quarantine_workflow, are not physical tables.

Every full-diagram table shows all its columns. PK means primary key. FK means declared foreign key. UK means unique constraint. ! marks a required column. Composite keys and unique constraints apply to their fields together.

Source: submitted schema and migrations 001 through 006 at commit 4a8ad8af275d8ee039873b4b2036e0f09c182b16. An isolated PostgreSQL 18 reconstruction matched all 164 columns in the retained PostgreSQL 16 workflow checkpoint and added the eight lifecycle columns from migration 003. No live RDS query or database change was made for this diagram.

transactions.batch_id is lineage, not a declared foreign key. candidate_account_id and user subject fields also have no declared SQL FK. Solid connectors represent only the database foreign keys.

risk_reviews is identified by risk_event_id. risk_review_history uses the composite primary key (risk_event_id, version). Reliable source identity has a partial unique index on (source_system, account_id, source_transaction_id) when the source reference is present. Transaction fingerprints are not globally unique.

Cognito owns application identities. S3 holds source files, versioned evidence payloads and reports. DynamoDB holds current cloud risk projections and alert claims. The alerts SQL table remains for the legacy/direct path. No fictional SQL users, notifications or rule-detail tables have been added.

Frozen JSON assessments and outboxes preserve deliberate audit and recovery snapshots. The diagram documents the physical implementation and does not claim that every stored snapshot is a fully normalised relation.

Open the exact submission source